$de \equiv 1 \pmod{(p-1)(q-1)}$

Problem 1

(a) $\gcd(135, 59) = 1$

x	y	$\text{rem}(x,y) = x - q \cdot y$
135	59	$17 = 135 - 2 \cdot 59$
59	17	$8 = 59 - 3 \cdot 17$
		$= 59 - 3(135 - 2 \cdot 59) = 59 - 3 \cdot 135 + 6 \cdot 59$
		$= -3 \cdot 135 + 7 \cdot 59$
17	8	$1 = 17 - 2 \cdot 8$
		$= 135 - 2 \cdot 59 - 2(-3 \cdot 135 + 7 \cdot 59)$
		$= \boxed{7 \cdot 135 - 16 \cdot 59}$
8	1	$0 =$

$s = 7, \quad t = -16$

(b) $\gcd(135, 59) = 1$

$7 \cdot 135 - 16 \cdot 59 = 1$

$0 + (-16) \cdot 59 \equiv 1 \pmod{135}$

$(-16) \cdot 59 \equiv 1 \pmod{135}$

(c) $17 \cdot x \equiv 1 \pmod{31}$ $\quad \phi(n) =$ if n prime, $n-1$

$\gcd(17, 31) = 1$

$17^{\phi(31)} \equiv 1 \pmod{31}$

$17^{30} \equiv 1 \pmod{31}$

$17^{29} \cdot 17 \equiv 1 \pmod{31}$

$17^{29} \equiv 17^{16} \times 17^{8} \times 17^{4} \times 17 \pmod{31}$

$\equiv 14 \times 18 \times 7 \times 17 \pmod{31}$

$\equiv 11 \pmod{31}$

$x = 11$

(d) by Euler's theorem, $\gcd(34, 83) = 1$

$$34^{82} \equiv 1 \pmod{83}$$

$$\text{rem}(34^{82^{82^{98}}}, 83) \equiv 34^{82^{82^{98}}} \pmod{83}$$

$$\equiv 34^{2+82(1003)} \pmod{83}$$

$$\equiv 34^2 \times (34^{82})^{1003} \pmod{83}$$

$$= 77 \times 1$$

$$\equiv 77 \pmod{83} \quad \square$$

Problem 2

(a) If $a|b$ then $\forall c, a|bc$

$a|b \Rightarrow b = ak$

$bc = akc$

$bc = a(kc)$

since kc is just an integer, let $kc = m$

$bc = am$

$a \mid bc$ $\square$

(b) If $a|b$ and $a|c$, then $a|sb+tc$

$a|b \Rightarrow ak_1 = b$

$a|c \Rightarrow ak_2 = c$

$sb + tc = s(ak_1) + t(ak_2)$

$= a(sk_1 + tk_2)$

since $sb+tc$ is multiple of a, $a|sb+tc$ $\square$

(c) $\forall c, a|b \Leftrightarrow ca|cb$

$a|b \Leftrightarrow ak = b \Leftrightarrow akc = bc$

$\Leftrightarrow ca(k) = cb$

$\Leftrightarrow ca|cb$ □

(d) $\gcd(ka, kb) = k\gcd(a, b)$

$\gcd(a, b) = sa + tb = g$

$k(sa+tb) = kg$

$(ks)a + (kt)b = kg$

since ks & kt are integers, kg is $\gcd(ka, kb)$

$\gcd(ka, kb) = kg = k\gcd(a, b)$ □

---

problem 3

(a) $x^2 = y^2 \pmod p \Leftrightarrow x^2 - y^2 \equiv 0 \pmod p$

$\Leftrightarrow (x+y)(x-y) \equiv 0 \pmod p$

since p is a prime & by Euclid's Lemma

$x+y \equiv 0 \pmod p$ or $x-y \equiv 0 \pmod p$

$x \equiv -y \pmod p$ (or) $x \equiv y \pmod p$ □

---

n is a square modulo $p \Rightarrow n \equiv x^2 \pmod p$

(b) By Fermat's little Theorem, $x^{p-1} \equiv 1 \pmod p$

$(x^2)^{\frac{p-1}{2}} = 1 \pmod p$

$n^{\frac{p-1}{2}} \equiv 1 \pmod p$ □

mostly we check up to $p-1$ to find every square modulo p

(c) $4k+3 \equiv 3 \pmod 4$

$n \equiv x^2 \pmod{4k+3}$

By Euler's Criterion, $n^{2k+1} \equiv 1 \pmod p$

$n^{2k+2} \equiv n \pmod p$

$(n^{k+1})^2 \equiv n \pmod p$

$x = n^{k+1} \pmod p$ □

Problem 4

for any prime, p, and integer, $k \geq 1$, $\phi(p^k) = p^k - p^{k-1}$

$\phi(p^k)$ = Total number − multiples of p → by $0p, 1p, 2p, \ldots, p^k - p$

$p(p^{k-1} - 1)$

n multi

By counting intervals in an interval (both inclusive) for multiples of p

$$[a, b] = b - a + 1$$

$$[0, p^{k-1} - 1] = p^{k-1} - 1 + 1$$

$$= p^{k-1}$$

$\phi(p^k)$ = Total number − multiple of p

$$= p^k - p^{k-1}$$

Problem 5

(a) Pf: by induction

$P(n)$ every number on board after n moves is either x, y, or a positive divisor of $\gcd(x, y)$.

Base case

$P(0)$: nothing new number ✓

Inductive step

Assume $P(n)$ is true & show $P(n) \Rightarrow P(n+1)$

a player write a new number, d which are common divisor of a & c

by $P(n)$: we know both a & c are x, y, or a positive divisor of $\gcd(x, y)$

by transitivity of divisibility, d must a positive divisor of $\gcd(x, y)$

(b) Proof by Contradiction

suppose there was d, a positive divisor of $\gcd(x,y)$ that is not on the board when the game end

since $d \mid \gcd(x,y) \Rightarrow d \mid x$ and $d \mid y$

so we can write down d on board but the game end b.c there is no move

Contrdiction

$\therefore$ every postive divisor of $\gcd(x,y)$ is on board at the end of the game □

(c) strategy for winning: go first

the number that can write on board always be odd $(\gcd + 2*n)$ where n is one side divisor of gcd e.g $(\gcd(x,y) = 6, 6\times1, 2\times3,) \ 6, 2$

and mirror every move of your opponent.

---

problem 6

(a) proof: by contradiction

let set F of all prime number is finite, $F = \{p_1, p_2, \cdots, p_k\}$

$$n = p_1 p_2 \cdots p_k + 1$$

n can't be divided by primes $p_1, \cdots, p_k$, b.c it leave remainder 1,

n is a prime which is contrdiction to F

thetore there initinite number of prime □

(b) every prime except 2 is odd

since 4 is dividing odd prime, rem $(n, 4)$ is either 3, 1

(c) $n = 3 \pmod 4$, then n has a prime factor $p \equiv 3 \pmod 4$

Proof by contradiction

non of the prime factors of $n \not\equiv 3 \pmod 4$

since n is odd (rem = 3) & by 6(b), all prime factor must be $p \equiv 1 \pmod 4$

by fundmental theorem of arithmetic

$$n = p_1 \cdot p_2 \cdots p_k$$

$$n \equiv 1 \times 1 \times \cdots \times 1 \pmod 4$$

which is contrdaction □

(d) Proof by contradiction

Let F is a finite set that is $\{P_1, \ldots, P_k\}$

define $n = 4 P_1 P_2 \cdots P_k - 1$

n is not divisible by all prime in F so n is a prmie

divide n by 4 will leave remeinde 3 which

$$n \equiv 3 \pmod 4$$

But n is not inside F This is a contradiction ∎